

Numerical Techniques 2026–2027

0. Welcome

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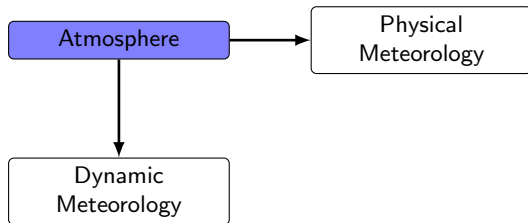
Postgraduate Studies in Weather and Climate Modeling

Ghent University

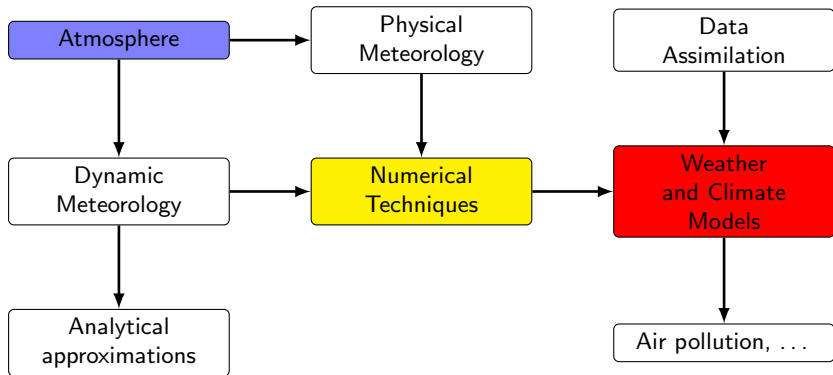
- Welcome
- Context: why numerical techniques
- Objectives of this course
- Course material
- Practical information

Why numerical techniques?

Atmosphere



Why numerical techniques?



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- Don't be frightened by code
... but it's no course on programming either!

- AI-driven weather models are performing better and better, sometimes beating physics-based models on specific scores, often at a fraction of the computational cost
- [*personal assessment!*] This could make physics-based models less relevant, especially for operational weather forecasting
- Nevertheless,
 - ▶ Whether extreme events are forecasted well by an AI model is an open question
 - ▶ Physics-based models remain (for now) important for training AI models
 - ▶ There is value in scientific understanding
 - ▶ Many aspects treated in this course are also relevant for AI models, e.g. time-stepping, stability, scalability, etc.

date	16h00	17h30
21/09	Introduction, Stability	(optional) Practicum Python basics
28/09	Time discretization	Practicum
05/10	Space discretization	
12/10	(no lesson)	
19/10	Spectral methods	Practicum
26/10	Nonlinearity	Practicum
02/11	(no lesson)	
09/11	Semi-Implicit and Semi-Lagrangian models	Project assignment
16/11	(no lesson)	
23/11	Guest lecture on AI	
30/11	Parallel computing	Project support session
07/12	Project support session	
08/12	Guest lecture on a real weather model	(morning)
14/12	Student project presentations	

- Slides will appear on Ufora
- All material (slides sources, Jupyter notebooks) available on https://github.com/ddegrauwe/ugent_numtech
- References:
 - ▶ *Numerical Methods for Wave Equations in Geophysical Fluid Dynamics*, Dale R. Durran, Springer, 1999, ISBN 0-387-98376-7.
 - ▶ *Chebyshev and Fourier Spectral Methods*, John P. Boyd, Springer, 2001, ISBN 978-3-540-51487-9.
- Some papers (depending on project)

- (Check Ufora for modifications to time schedule)
- Practical sessions
 - ▶ Software needed: python (numpy, ipympl, matplotlib), Jupyter notebooks
 - ▶ We will use High-Performance Computing (HPC) infrastructure of UGent: you should already have an account (?)
 - ▶ access through browser via <https://login.hpc.ugent.be>
 - ▶ or you can just install Jupyter on your laptop
- Evaluation: student project (2/3/4 persons) on simple model
 - ▶ presentation for other students
 - ▶ (small) report

Questions?